Practical 8

	Present state				Next state			
	A	B	C	D	A (t+1)	B (t+1)	C (t+1)	D (t+1)
7	0	1	1	1	0	0	1	1
3	0	0	1	1	1	0	1	1
11	1	0	1	1	1	0	0	1
9	1	0	0	1	0	0	1	0
2	0	0	1	0	0	0	0	1
1	0	0	0	1	0	0	0	0
0	0	0	0	0	0	1	1	0
6	0	1	1	0	1	0	0	0
8	1	0	0	0	1	1	1	1
15	1	1	1	1	0	1	0	0
4	0	1	0	0	0	1	1	1

	J_A	K_A	J_B	K_B	J_C	K_C	J_D	K_D
7	0	x	x	1	x	0	x	0
3	1	x	0	x	x	0	x	0
11	x	0	0	x	x	1	x	0
9	x	1	0	x	1	x	x	1
2	0	x	0	x	x	1	1	x
1	0	x	0	x	0	x	x	1
0	0	x	1	x	1	x	0	x
6	1	x	x	1	x	1	0	x
8	x	0	1	x	1	x	1	x
15	x	1	x	0	x	1	x	1
4	0	x	x	0	1	x	1	x

$J_A =$

	$C'D'$	$C'D$	CD	CD'
$A'B'$	0	0	0	0
$A'B$	0	x	0	1
AB	x	x	x	x
AB'	x	x	x	x

$J_A = B'CD + BCD'$

$C(B'D + BD')$

$C(B \oplus D)$

$K_A =$

	$C'D'$	$C'D$	CD	CD'
$A'B'$	x	x	x	x
$A'B$	x	x	x	x
AB	x	x	x	x
AB'	0	1	0	x

$K_A = C'D + B$

$J_B =$

	$C'D'$	$C'D$	CD	CD'
$A'B'$	1	0	0	0
$A'B$	x	x	x	x
AB	x	x	x	x
AB'	1	0	0	x

$J_B = C'D'$

$K_B =$

1	x	x	x
0	x	x	1
x	x	0	x
x	x	x	x

$K_B = A'C$

$K_C =$

X	X	0	1
X	X	0	1
X	X	1	X
X	X	1	X

$K_C = A + D'$

$J_C =$

1	0	X	X
1	X	X	X
X	X	X	X
X	1	X	X

$J_C = D' + A$

$K_D =$

X	1	0	X
X	X	0	X
X	X	1	X
X	1	0	X

$K_D = C' + AB$

$J_D =$

0	X	X	1
1	X	X	0
X	X	X	X
1	X	X	X

$J_D = A + B \oplus C$